

Ex.No: 8

Exercises On Redux

Date:

Aim:

To implement React Programs on REDUX

Problem:

Shopping Cart: Store: cart (items with id, name, price, qty); Actions: ADD_ITEM, REMOVE_ITEM, UPDATE_QTY. Calculate total price from Redux state.

Use redux-persist to keep auth or cart in localStorage.

Code:

[store.js](#):

```
import { configureStore, createSlice } from "@reduxjs/toolkit";  
  
import storage from "redux-persist/lib/storage";  
  
import { persistReducer, persistStore } from "redux-persist";  
  
import { combineReducers } from "redux";  
  
  
const cartSlice = createSlice({  
  name: "cart",  
  initialState: { items: [] },  
  reducers: {
```

```

ADD_ITEM: (state, action) => {
  const item = action.payload;
  const existing = state.items.find(i => i.id === item.id);
  if (existing) {
    existing.qty += item.qty;
  } else {
    state.items.push(item);
  }
},

REMOVE_ITEM: (state, action) => {
  state.items = state.items.filter(i => i.id !== action.payload);
},

UPDATE_QTY: (state, action) => {
  const { id, qty } = action.payload;
  const existing = state.items.find(i => i.id === id);
  if (existing) {
    existing.qty = qty;
  }
}

});

export const { ADD_ITEM, REMOVE_ITEM, UPDATE_QTY } = cartSlice.actions;

// persist config

```

```
const persistConfig = {  
  key: "root",  
  storage,  
};
```

```
const rootReducer = combineReducers({  
  cart: cartSlice.reducer  
});
```

```
const persistedReducer = persistReducer(persistConfig, rootReducer);
```

```
export const store = configureStore({  
  reducer: persistedReducer  
});
```

```
export const persistor = persistStore(store);
```

App.jsx:

```
import { useDispatch, useSelector } from "react-redux";  
import { ADD_ITEM, REMOVE_ITEM, UPDATE_QTY } from "../store";  
  
export default function App() {  
  const dispatch = useDispatch();  
  const cartItems = useSelector((state) => state.cart.items);
```

```
const totalPrice = cartItems.reduce((sum, item) => sum + item.price * item.qty, 0);
```

```
const addSampleItem = () => {  
  dispatch(ADD_ITEM({ id: 1, name: "Laptop", price: 50000, qty: 1 }));  
};
```

```
const updateQty = (id, qty) => {  
  dispatch(UPDATE_QTY({ id, qty }));  
};
```

```
return (  
  <div style={{ padding: 20 }}>  
    <h1> Shopping Cart</h1>  
    <button onClick={addSampleItem}>Add Laptop</button>  
  
    <ul>  
      {cartItems.map((item) => (  
        <li key={item.id}>  
          {item.name} - ₹ {item.price} × {item.qty} = ₹ {item.price * item.qty}  
          <button onClick={() => updateQty(item.id, item.qty + 1)}>+</button>  
          <button onClick={() => updateQty(item.id, item.qty - 1)}>-</button>  
          <button onClick={() => dispatch(REMOVE_ITEM(item.id))}>Remove</button>  
        </li>  
      ))}  
    </ul>  
)
```

```
    <h2>Total: ₹{totalPrice}</h2>
  </div>

);
}
```

Main.jsx:

```
import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App.jsx";
import { Provider } from "react-redux";
import { PersistGate } from "redux-persist/integration/react";
import { store, persistor } from "./store.js";

ReactDOM.createRoot(document.getElementById("root")).render(
  <Provider store={store}>
    <PersistGate loading={null} persistor={persistor}>
      <App />
    </PersistGate>
  </Provider>
);
```

Output:

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Shopping Cart

Add Laptop

• Laptop - ₹50000 × 1 = ₹50000

Total: ₹50000

Problem Analysis (10)	Coding & Implementation (10)	Time Management (10)	Viva (10)	Total (40)

Result:

Thus the program on redux has been executed successfully and the output has been verified.